



Autodesk Delivers a World Beyond Files Using Amazon DynamoDB

Learn how Autodesk helps the AEC industry realize better business outcomes with accessible data using Amazon DynamoDB.

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80+ million

rows-per-minute writing speed achieved



100,000 Revit documents

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Increased data accessibility

to drive better decisions



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Reduced rework



Overview

Software company [Autodesk](#) is imagining a world beyond files for the industries it serves. In the architecture, engineering, and construction (AEC) industry, companies store information for projects in large files that need to be accessed and shared between various teams. However, sharing monolithic files across teams can create bottlenecks, complicate data access, and raise versioning issues. Driven by the need to collaborate efficiently and access information quickly, the AEC industry is shifting from traditional file-based systems to data management in the cloud.

Autodesk wants to help its customers work more efficiently and intelligently by making design data more accessible and transparent through breaking down these monolithic files into smaller chunks of data hosted in the cloud. It is using Amazon Web Services (AWS) to create its AEC Data Model, a data management system designed to move teams from linear, file-based workflows to object-based, collaborative workflows.



Autodesk offers a design and make solution for the manufacturing, media, and AEC industries, including Autodesk Revit, one of the company's building information modeling (BIM) solutions. BIM is the process of creating and managing information on a built asset throughout the entire life cycle of that project. The kind of building projects that Revit is used for are incredibly complex and have hundreds to thousands of stakeholders. A single change to the data can have huge impacts. For example, making a minor change to the size or type of material for an asset in the file can have a ripple effect across the project and increase budgets by hundreds of thousands of dollars. In fact, rework can make up 11–12 percent of the average project's budget.

Autodesk has been working to create a solution to break down monolithic BIM files into granular data so that everyone on a project team can access all essential data in real time to collaborate more easily and effectively. It began creating a cloud-based data model for the manufacturing vertical, but it faced challenges recreating a similar solution for the AEC industry because of the complexity and amount of data. Consisting of dynamic 3D models of all the data associated with a project and capturing all physical and functional characteristics and relationships for buildings and infrastructure, AEC projects can have up to 100 times as much data as manufacturing models. "Supporting AEC workloads is not an easy task," says Srinivasan Jayaraman, software architect for product data at Autodesk. "We don't want to reinvent the wheel but rather use a managed service from AWS so that we can build on it and deliver value to our customers as quickly as possible."

In 2022, Autodesk began to unlock the data capabilities it needed using [Amazon DynamoDB](#), a serverless, NoSQL, fully managed database with single-digit millisecond performance at any scale. "Amazon DynamoDB is fully managed by AWS, and it has high availability, global replication, and high throughput," says Jayaraman. "It's also simple to build apps on top of it, and the support that we got from AWS was another key factor for choosing it."

In August 2022, Autodesk and AWS met for a 2-day workshop, followed by 9 months of continuous engagement to implement Amazon DynamoDB into the AEC Data Model. In mid-2023, Autodesk began a private beta for the AEC Data Model, and in November 2023, it opened a public beta, in which around 600 customers are participating as of March 2024.

Solution | Writing More than 80 Million Rows in 1 Minute Using Amazon DynamoDB

The AEC Data Model API unlocks granular data that has historically been trapped in Revit files and makes it more accessible and useful to customers. It makes available information such as asset type, material, and performance, warranty, and cost—a contractor could use the API to find the number and type of windows in a



building and then calculate the cost based on the dimensions, all without needing to open the design file directly. It also lays the foundation for building automations that eliminate or reduce manual processes for customers. For example, the number of steel beams in a model could be queried using an AI assistant rather than counted individually by opening up the model.

Large Revit projects can contain up to one million elements—individual assets of a building plan, such as windows or doors, that can be separated into granular data elements. To ingest this data into the AEC Data Model, Autodesk uses Amazon DynamoDB Streams, a change data capture capability in Amazon DynamoDB, and [Amazon Kinesis Data Streams](#), a serverless streaming data service.

These services ingest elements into Asset Graph, another layer of the AEC Data Model solution, and the data is aggregated using [Amazon Managed Service for Apache Flink](#), a service to set up and integrate data sources and destinations with minimal code and continuously process data with subsecond latencies. Within Asset Graph, Autodesk also built a system to index and search the data. The AEC Data Model API then takes the results from Asset Graph and returns them to customers.

Amazon DynamoDB serves as the single source of truth for data in the AEC Data Model. Autodesk forecasts that the AEC Data Model will have daily usage of 100,000 Revit documents totaling 7 billion elements, 90 billion parameters, and 3 billion relationships. Using Amazon DynamoDB, the solution can scale to handle these high volumes of data; during testing, it read and wrote 81 million rows in one minute.

Using the AEC Data Model, Autodesk customers can accelerate projects because multiple teams can access BIM data at the same time. The data is kept up to date, so versioning is no longer an issue either. This means teams can view and react to changes to a model in real time, avoiding rework and delays. Autodesk has created connectors on top of the AEC Data Model, such as a GraphQL API to deliver data results, and customers are also building their own connectors to other applications. “Today, our customers spend a lot of time in upload queues, waiting for their changes to sync. With the AEC Data Model API, when someone makes a change in the model, there will be immediate visibility into potential upstream and downstream effects,” says Virginia Senf, director of AEC industry strategy at Autodesk. “The AEC Data Model delivers increased visibility into changes as they happen so that the right people who are impacted by that change know about it at the right time.”

Outcome | Creating a Unified Data Solution Beyond Files Using AWS

The AEC Data Model API will be generally available for Revit workloads in 2024. Subsequently, Autodesk plans to implement data models for its other products as well. “With the AEC Data Model, we’re solving the root of the problem,” says Senf. “Using AWS, we’re building a unified solution that we can use to help our customers maximize the value of their data and realize better outcomes for their business and the world. We can scale it moving forward as data volumes continue to explode.”

About Autodesk

Autodesk is a global leader in software for designers, engineers, builders, and creators. Autodesk’s Design and Make Platform empowers customers with the technology to create the world around us.



AWS Services Used

Amazon DynamoDB

Amazon DynamoDB is a serverless, NoSQL, fully managed database service with single-digit millisecond response times at any scale, enabling you to develop and run modern applications while only paying for what you use.

[Learn more »](#)

Amazon Kinesis Data Streams

Amazon Kinesis Data Streams is a serverless streaming data service that makes it easy to capture, process, and store data streams at any scale.

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Amazon Managed Service for Apache Flink

Set up and integrate data sources and destinations with minimal code, continuously process data with subsecond latencies, and respond to events in real time.

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